

A Natural L_r -Class Obstruction For Question 2.12 Of arXiv:2402.19152

Automatic math-discovery smoke test

June 8, 2026

Source Problem

The source paper is:

E. García-Sánchez, D. H. Leung, M. A. Taylor, P. Tradacete,

Banach lattices with upper p -estimates: free and injective objects, arXiv:2402.19152, Functional Analysis.

The relevant open question appears as Question 2.12 on page 11 of the arXiv PDF. A screenshot crop of the paper is included below.

Thus, $\rho(J) \geq \rho c(J)$.

□

The above construction of $\text{FBL}^C[E]$ motivates the following question.

Question 2.12. It is well-known that the properties of $\text{FBL}^{(p)}[E]$ depend heavily on whether p is finite or infinite. Can the Banach lattices $\text{FBL}^C[E]$ be used to interpolate such properties? For example, if (e_k) is the unit vector basis of ℓ^2 , then by [28, Propositions 5.14 and 6.4] $(|\delta_{e_k}|)$ gives a copy of ℓ^1 in $\text{FBL}^{(p)}[\ell^2]$ when $p < \infty$ and a copy of ℓ^2 when $p = \infty$. Can we find for each $r \in (1, 2)$ a class $\mathcal{C}_r$ so that $(|\delta_{e_k}|)$ behaves like ℓ^r in $\text{FBL}^{\mathcal{C}_r}[\ell^2]$?

2.4. Injective homomorphisms need not extend to the norm completion. In Section 2.2 we defined $\text{FBL}^C[E]$ as the completion of $(\text{FBL}[E], \rho)$. Clearly there is a continuous lattice ho-

The final part of the question asks whether, for each $r \in (1, 2)$, one can find a class $\mathcal{C}_r$ such that the sequence $(|\delta_{e_k}|)$, where (e_k) is the unit vector basis of ℓ^2 , behaves like the unit vector basis of ℓ^r in $\text{FBL}^{\mathcal{C}_r}[\ell^2]$.

Result Type

This packet is a **partial result**. It does not answer the full existence question for arbitrary classes $\mathcal{C}_r$. It rules out the most direct single-space candidate suggested by the preceding rearrangement-invariant construction: $\mathcal{C}_r = \{L_r[0, 1]\}$.

Main Claim

Theorem 1. Let $1 \leq q \leq 2$, let $\mathcal{C}_q = \{L_q[0, 1]\}$, and let

$$u_k = |\delta_{e_k}| \in \text{FBL}^{\mathcal{C}_q}[\ell^2],$$

where (e_k) is the unit vector basis of ℓ^2 . Then for every finitely supported family of nonnegative scalars (a_k) ,

$$\left\| \sum_k a_k u_k \right\|_{\text{FBL}^{\mathcal{C}_q}[\ell^2]} = \sum_k a_k.$$

Proof. Let

$$f_a = \sum_k a_k |\delta_{e_k}| \in \text{FVL}[\ell^2],$$

where $a_k \geq 0$ and only finitely many a_k are nonzero. By the definition of the norm associated to a class of Banach lattices,

$$\rho_{\mathcal{C}_q}(f_a) = \sup \left\{ \|\widehat{T}f_a\|_{L_q} : T : \ell^2 \rightarrow L_q[0, 1] \text{ is a linear contraction} \right\}.$$

First, let $T : \ell^2 \rightarrow L_q[0, 1]$ be any contraction. Since the lattice homomorphic extension satisfies $\widehat{T}(|\delta_{e_k}|) = |Te_k|$, the triangle inequality gives

$$\|\widehat{T}f_a\|_{L_q} = \left\| \sum_k a_k |Te_k| \right\|_{L_q} \leq \sum_k a_k \|Te_k\|_{L_q} \leq \sum_k a_k.$$

Taking the supremum over all contractions T gives

$$\rho_{\mathcal{C}_q}(f_a) \leq \sum_k a_k.$$

For the reverse inequality, suppose the support of (a_k) is contained in $\{1, \dots, n\}$. Let $r_1, \dots, r_n$ be the first n Rademacher functions on $[0, 1]$. Define

$$T : \ell^2 \rightarrow L_q[0, 1], \quad T \left(\sum_{k \geq 1} x_k e_k \right) = \sum_{k=1}^n x_k r_k.$$

Since $[0, 1]$ has measure 1 and $q \leq 2$,

$$\|Tx\|_{L_q} \leq \|Tx\|_{L_2} = \left(\sum_{k=1}^n |x_k|^2 \right)^{1/2} \leq \|x\|_{\ell^2}.$$

Thus T is a contraction. Moreover, $|r_k| = 1$ almost everywhere, so

$$\widehat{T}f_a = \sum_{k=1}^n a_k |r_k| = \sum_{k=1}^n a_k$$

as a constant function. Hence

$$\rho_{\mathcal{C}_q}(f_a) \geq \|\widehat{T}f_a\|_{L_q} = \sum_{k=1}^n a_k.$$

Together with the upper estimate, this proves the equality. $\square$

Corollary 1. *For every $r \in (1, 2)$, the natural candidate $\mathcal{C}_r = \{L_r[0, 1]\}$ does not make $(|\delta_{e_k}|)$ behave like the unit vector basis of ℓ^r in $\text{FBL}^{\mathcal{C}_r}[\ell^2]$.*

Proof. Apply the theorem with $q = r$ and $a_1 = \dots = a_n = 1$. Then

$$\left\| \sum_{k=1}^n |\delta_{e_k}| \right\|_{\text{FBL}^{\{L_r\}}[\ell^2]} = n.$$

If the sequence were equivalent to the unit vector basis of ℓ^r , there would be a constant C independent of n such that this norm were at most $Cn^{1/r}$. This is impossible because $n/n^{1/r} = n^{1-1/r} \rightarrow \infty$. $\square$

Verification Notes

The argument is proof-theoretic and uses only the paper’s definition of $\rho_{\mathcal{C}}$, the lattice homomorphism extension property, and the standard Rademacher functions. The accompanying script `check_positive_growth.py` checks the finite Walsh model numerically for sample values of $q \in (1, 2)$. The script is not needed for the proof; it is included as a small sanity check of the contraction and growth mechanism.

Limitations And Novelty Check

This is not a full solution of Question 2.12. It only says that the obvious rearrangement-invariant candidate $\{L_r[0, 1]\}$ cannot provide the desired interpolation. More exotic classes $\mathcal{C}_r$ remain open.

A web search on June 8, 2026 for the exact interpolation question and nearby phrases located the source paper and publication mirrors, but no later paper explicitly answering Question 2.12. This packet should be treated as a candidate partial result for human review.

Human Review Recommendation

The proof is short and appears likely valid. The key points to check are: the normalization of $L_q[0, 1]$, the use of the universal norm $\rho_{\mathcal{C}_q}$, and the fact that the candidate result is only a partial negative result for the natural single-space class.